

Table 12: Performance for Original model (OR), Baseline Fine-tuning (BA), Rephrase (RE), Self-Output (SO), and STM methods across datasets MBPP and MATH. Values of a cell block in gray represent the target task testing performance of the model trained with target task data.

MODEL	TARGET TASK	METHOD	MBPP	MATH	ARC	GSM8K	BIRD
GEMMA 2 IT 2B	-	OR	0.5106	0.2810	0.7474	0.5845	0.1389
		BA	0.3995 ↓	0.2290 ↓	0.7159 ↓	0.1926 ↓	0.0514 ↓
		RE	0.4841 ↓	0.2890 ↑	0.7056 ↓	0.4822 ↓	0.1375 ↓
		SO	0.5344 ↑	0.2920 ↑	0.5845 ↓	0.5239 ↓	0.1395 ↑
	MBPP	STM	0.5106	0.2980 ↑	0.7483 ↑	0.5641 ↓	0.1375 ↓
		BA	0.4577 ↓	0.2170 ↓	0.2952 ↓	0.1903 ↓	0.1245 ↓
		RE	0.4683 ↓	0.2000 ↓	0.7338 ↓	0.3700 ↓	0.1258 ↓
		SO	0.5000 ↓	0.3090 ↑	0.7295 ↓	0.5701 ↓	0.1389 -
	MATH	STM	0.4947 ↓	0.3030 ↑	0.7534 ↑	0.5542 ↓	0.1330 ↓
		OR	0.4868	0.1720	0.6408	0.3169	0.1284
	-	BA	0.4550 ↓	0.1220 ↓	0.6826 ↑	0.3685 ↑	0.1617 ↑
		RE	0.4524 ↓	0.1910 ↑	0.6775 ↑	0.3889 ↑	0.1649 ↑
MISTRAL 7B INSTRUCT	MBPP	SO	0.4709 ↓	0.1900 ↑	0.6903 ↑	0.2252 ↓	0.1375 ↑
		STM	0.4788 ↓	0.1900 ↑	0.6101 ↑	0.4905 ↑	0.1239 ↓
	MATH	BA	0.4709 ↓	0.1670 ↓	0.3498 ↓	0.000 ↓	0.1851 ↑
		RE	0.4974 ↑	0.1650 ↓	0.7355 ↑	0.2714 ↓	0.1799 ↑
		SO	0.4683 ↓	0.1940 ↑	0.7312 ↑	0.2396 ↓	0.1447 ↑
		STM	0.4656 ↓	0.1890 ↑	0.6510 ↑	0.4541 ↑	0.1167 ↓
LLAMA 3 8B INSTRUCT	-	OR	0.5926	0.3140	0.7816	0.7255	0.2053
		BA	0.5794 ↓	0.2330 ↓	0.5913 ↓	0.1850 ↓	0.1760 ↓
		RE	0.5406 ↓	0.3000 ↓	0.7543 ↓	0.7240 ↓	0.1916 ↓
		SO	0.6164 ↑	0.3340 ↑	0.7918 ↑	0.7839 ↑	0.2001 ↓
	MBPP	STM	0.6111 ↑	0.3200 ↑	0.7892 ↑	0.7543 ↑	0.2066 ↑
		BA	0.5873 ↓	0.2580 ↓	0.4863 ↓	0.6793 ↓	0.1942 ↓
		RE	0.6032 ↑	0.3290 ↑	0.7850 ↑	0.7862 ↑	0.1890 ↓
		SO	0.6005 ↑	0.3440 ↑	0.8012 ↑	0.7801 ↑	0.1988 ↓
	MATH	STM	0.6164 ↑	0.3340 ↑	0.7509 ↓	0.7506 ↑	0.2027 ↓

D.6 Convergence curve with Gemma 2 IT 2B

To validate the model size effect in terms of model convergence, we also observe the training procedure of Gemma 2 IT 2B. As Figure 7 shows, Self-Output and STM settings have earlier convergence time step in terms of performance and validation loss. And maintain a relative low training loss curve compared to Ground Truth data training as the same trend we see in Llama 3 8B Instruct’s case.

E High perplexity tokens filtered by STM

We have noticed that the Self-output responses consist with low surprisal values as highly predictable token sequences. It is also worthwhile to note that the contents of high perplexity tokens between Ground Truth responses and Self-ouotput responses are different. In Math dataset, we simply categorize the tokens in model responses as **Numbers**, **Symbols** and **Words**. Numbers refers to the tokens which are numeric words. Symbols refers to tokens that imply a non-numeric symbols or math word in latex, such as $+$, $-$, $*$, $/$. Words refers to tokens consisting of alphabets only. As Figure 8, Figure 9, and Figure 10 shows, in terms of all the models we have tested, the proportion of number tokens with high perplexity drop drastically in Self-output responses, while the overall distribution of the three categories in both Ground Truth responses and Self-Output responses are almost the same. It implies that the number tokens in Ground Truth dataset are less predictable by models and harder to learn. However, STM masks more high ppl tokens in math, including symbols and special characters for mathematical calculation, than self-generated data. But the performance of

Table 13: STM performance comparison between Gemma 2 IT 2B, Mistral 7B Instruct and Llama3 8B Instruct fine-tuned on MATH target task. The percentage indicates the number of filtered tokens from the training data.

MODEL	MATH	MBPP	GSM8K	ARC	BIRD
<i>Gemma 2 IT 2B</i>					
BASELINE FINE-TUNING	21.7	45.8	19.0	29.5	5.1
$STM_{\tau=1000}$ (2.5%)	23.3	48.9	24.9	25.4	12.8
$STM_{\tau=25}$ (9.3%)	27.0	48.2	49.7	72.9	13.1
$STM_{\tau=2.5}$ (23.8%)	30.3	49.5	55.4	75.3	13.3
<i>Mistral 7B Instruct</i>					
BASELINE FINE-TUNING	16.7	47.1	0.0	35.0	16.2
$STM_{\tau=1500}$ (3.7%)	14.9	51.6	31.0	58.0	15.2
$STM_{\tau=30}$ (10.8%)	16.1	41.6	41.6	62.0	14.0
$STM_{\tau=3}$ (21.9%)	18.9	46.6	45.4	65.1	11.7
<i>Llama 3 8B Instruct</i>					
BASELINE FINE-TUNING	25.8	58.7	48.6	67.9	19.4
$STM_{\tau=1000}$ (1.7%)	27.7	59.2	69.3	71.2	19.4
$STM_{\tau=10}$ (11.0%)	33.0	61.1	77.6	75.6	19.5
$STM_{\tau=2.5}$ (22.2%)	33.4	61.6	75.1	75.1	20.3

Table 14: List of alternative token selection methodology supervised fine-tuned on MATH in Llama 3 8B Instruct. Here we only show the STM, DPF and STM+DPF version since Llama 3 8B Instruct does not have a feasible model size to implement the cross-scale setting for comparison.

LLAMA 3 8B INSTRUCT	FILTER	MATH	MBPP	GSM8K	ARC	BIRD
STM $\tau = 2.5$	22.2%	33.4	61.6	75.8	76.9	20.3
DPF $\tau = 0$	67.1%	23.9	57.1	48.1	70.1	18.9
$STM + DPF$ $\tau = 100$	70.9%	17.8	55.0	46.4	58.7	18.1

TI is still positive and better than baseline fine-tuning, showing that the model still improves without knowing those important key symbols but from the context without the masked words.

F Reproducibility

F.1 Training data cost

As we know there is efficiency difference of the data preparation between Self-Output, Rephrase, Selective Token Masking and Baseline Fine-tuning. To brief about the difference, Self-Output is the most resource-exhausted since it requires to generate N samples (usually set 32), validate the correctness of samples (including parsing out the answer, and match with gold label or pass test cases), and then do forward passes on those samples to pick the lowest perplexity sentence as training data. On the other hand, STM only requires a single forward pass on ground truth data to calculate perplexity of each token to do the masking in training process. As Table 17 shows, for a simple MBPP task, on a A100 GPU server to prepare for 374 instances (train) + 90 instances (validation) for $N=32$ takes about more than 18 gpu hours to complete the process of Self-Output to generate trainable data. While STM only requires 4 minutes to calculate the whole token perplexity.

F.2 Model training resource

We train Llama 3 8B Instruct, Mistral 7B Instruct and Gemma 2 IT 2B using two NVIDIA A100 Tensor Core GPU with VRAM 40GB for each GPU and additional RAM of 96GB. In terms of gpu

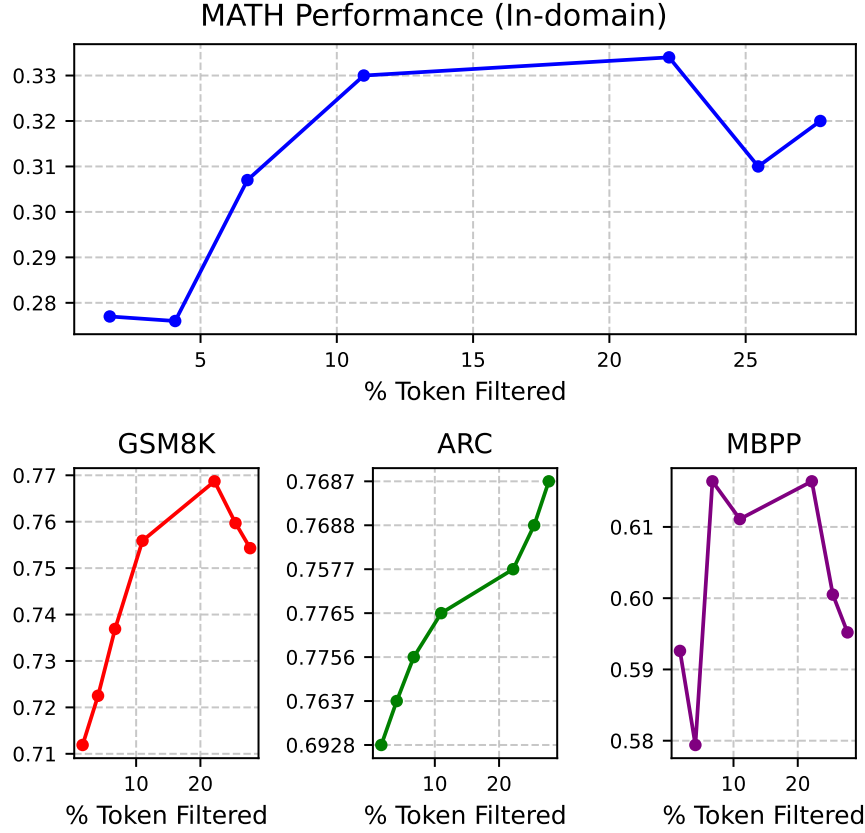


Figure 5: MATH SFT using STM method on Llama 3 8B Instruct on different levels of token filtering levels. The best in domain performance also matches peak performance on out of domain tasks : GSM8k, ARC, MBPP as well.

Table 15: Ablation and Scaling on STM threshold for Llama 3 8B Instruct model on MBPP target task.

LLAMA 3 8B INSTRUCT	BWT(%)	TI(%)
$STM_{\tau=2.5, high}$	-0.16	3.2
$STM_{\tau=2.5, random}$	-40.41	-9.84
$STM_{\tau=2.5, low}$	-33.01	-14.73
BASLINE FINE-TUNING	-14.1	-17.8
$STM_{\tau=1000}$ (1.7%)	-19.6	-11.8
$STM_{\tau=10}$ (11.0%)	-13.1	5.1
$STM_{\tau=2.5}$ (22.2%)	-0.3	6.4

hour, Our training experiments takes at most 2 hours for MBPP task, 4 to 6 hours for MATH task. As for the evaluation experiments for other non-target tasks, evaluation in ARC-Challenge takes 3.5 hours, evaluation in GSM8k takes 4.5 hours, and evaluation in BIRD task takes 7.5 hours.

F.3 Evaluation prompt

For each task task, we design different prompts or directly apply the prompts from the dataset. Since we use regular expression and LLM to parse out the answer we need to evaluate, so besides the task